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Relationships Between E-cigarette Use and Subsequent Cigarette Initiation Among Adolescents in the PATH Study: An Entropy Balancing Propensity Score Analysis

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Abstract

Objective: This study aimed to examine the relationship between electronic cigarette use and subsequent combustible cigarette use, controlling for confounding by using a propensity score method approach.

Methods: Data from the first three annual waves of the Population Assessment of Tobacco and Health study were analyzed (n = 6309). Participants were tobacco naïve at Wave 1, had used e-cigarettes exclusively (n = 414), used combustible cigarettes exclusively (n = 46), or not used any tobacco products (n = 5849) at Wave 2. We conducted entropy balancing propensity score analysis to examine the association between exclusive e-cigarette or cigarette initiation and subsequent cigarette use at Wave 3, adjusting for nonresponse bias, sampling bias, and confounding.

Results: Among tobacco naïve youth, exclusive e-cigarette use was associated with greater risk for subsequent combustible cigarette smoking initiation (OR = 3.42, 95% CI = (1.99, 5.93)) and past 30-day combustible cigarette use (OR = 2.88, 95% CI = (1.22, 6.86)) in the following year. However, the latter risk was comparatively lower than the risk if youth started with a combustible cigarette (OR = 25.79, 95% CI = (9.68, 68.72)). Results of sensitivity analyses indicated that estimated effects were robust to unmeasured confounding.

Conclusions: Use of e-cigarettes in tobacco-naïve youth is associated with increased risk of subsequent past 30-day combustible cigarette use but the risk is an order of magnitude higher if they start with a combustible cigarette.

Keywords

E-cigarette use; combustible cigarette use initiation; youth; entropy balancing; sensitivity analysis

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Compliance with Ethical Standard: This study is a secondary data analysis of the publicly available data from the Population Assessment of Tobacco and Health Study (PATH). All data have been de-identified (by Inter-University Consortium for Political and Social Research before releasing to researchers) and the current research involved no human participants and/or animals.

Informed consent: This study is a secondary data analysis of publicly available data. No informed consent form was concerned.

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Tobacco use, in particular combustible cigarette smoking, is the leading cause of preventable death in the United States (US Department of Health and Human Services, 2014). In the past decade, use of Electronic Nicotine Delivery Systems (ENDS, electronic cigarettes, or e-cigarettes), noncombustible alternatives to cigarettes, has been increasing in youth and young adults, sparking concerns that their use might lead to subsequent combustible cigarette smoking (Leventhal et al., 2015; Niaura et al., 2014; Rigotti, 2015). Since e-cigarette use entered the US market in 2007, the rate of past-30-day e-cigarette use among youth has grown substantially, from 0.6% in 2011 to 10.5% in 2019 among middle school students; and from 1.5% in 2011 to 27.5% in 2019 among high school students. This rate has surpassed the rate of conventional combustible cigarette use (2.3% and 5.8%), and e-cigarettes have become the most commonly used tobacco product among youth and young adults (Center for Disease Control and Prevention, 2019). E-cigarettes contain nicotine, which can cause addiction and perhaps lead to progression of use to more efficient, combustible tobacco products, especially conventional combustible cigarettes.

Abundant findings showed an association between e-cigarette use and subsequent uptake of combustible cigarettes among tobacco naïve youth (Aleyan et al., 2018; Best et al., 2018; Chan et al., 2021; Chyderiotis et al., 2020; Coleman et al., 2019; Glasser et al., 2019; Harlow et al., 2019; Khouja et al., 2020; Leventhal et al., 2015; Soneji et al., 2017; Spindle et al., 2017; Wills et al., 2017; Keller-Hamilton, et al, 2021). Evidence from cross-sectional studies has showed an association between e-cigarette use and higher odds of subsequent cigarette initiation. Recently, evidence predominately from US youth using longitudinal data also revealed similar patterns (Soneji et al., 2017; Wills et al., 2017). Additional studies reported from other countries including Canada (Aleyan et al., 2018), Scotland (Best et al., 2017), and France (Chyderiotis et al., 2020) suggest similar associations. A majority of these studies used school-based samples or community samples, and controlled for risk factors including demographics, psychological (e.g., depression symptoms, substance use, sensation seeking), and other factors (relationship status, family history of smoking, peer smoking). Limitations of these studies include small sample sizes and inadequate controls for potential confounder variables (Chan et al., 2021; Glasser et al., 2019; Khouja et al., 2020).

Addressing confounding in observational studies is challenging. Confounding distorts the association between the exposure and outcome variable in the presence of the third variable (confounder), and is of concern in almost all observational studies (VanderWeele, 2017). For example, confounding factors including self-selection, health history, family environments, and many other factors are associated with both e-cigarette exposure and subsequent combustible cigarette use (Soneji et al., 2017). Various confounders including sociodemographic, interpersonal, behavioral, and social environmental factors, have been inconsistently adjusted for in similar studies, predominantly by including confounders as covariates in a regression model. Ignoring or omitting any confounder in analyses poses a risk of biased estimates of the true association between exposure and an outcome, hence compromising causal interpretation (Greenland et al., 1999).

The current study addresses the confounding problem in observational studies within a causal inference framework. Potential outcome or counterfactual framework methods have gained popularity in social, behavioral, and health research, and have also become

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increasingly popular in tobacco research. The present study used data from the Population Assessment of Tobacco and Health (PATH) study, investigating the impact of e-cigarette use on subsequent combustible cigarette use among tobacco naive youths. We used propensity score methods to address two questions.

Question 1: *What would be the change in risk of subsequent combustible cigarette use had the user not used ENDS?*

Question 2: To compare with the risk in Question 1, *what would be the change in risk of subsequent combustible cigarette use had the user not used a combustible cigarette*?

Method

Data and Study Participants

The PATH Study is the largest contemporary longitudinal study that documents tobacco use in a nationally representative sample of youth and adults in the United States (N=45,971; including 13,651 youth and 32,320 adults). Participants of the youth survey included the youth (12 – 17 yrs.) and their parents, who were interviewed via a computer aided system annually starting from 2013. When youth turned 18, they were invited to participate in the adult survey. Study design and data collection has been reported in Hyland et al. (2017). We used the PATH data for public use which disguised actual age by grouped age (12 – 14, 15 – 17). Among the sample aged 12 – 15 yrs at Wave 1, 7595 completed Waves 1 – 3 of the youth survey, and at least 1413 were absent at Wave 2 or Wave 3. There were 3451 participants aged 16 – 17 who completed Waves 1 – 3 of the youth survey and at least 370 were absent at Wave 2 or Wave 3. Due to the fact that actual ages were not provided in the data for public use, we were not able to identify the ages of the remaining 971 youth who were between 15 – 17 and were absent at Wave 2 or Wave 3.

In the current study, we analyzed the first three waves of the PATH Study data (collected annually from 2013/14 to 2015/16) from a tobacco naïve youth sample aged between 12 and 15 years at Wave 1 (September, 2013 – December, 2014; n = 6309). Participants included Wave 2 E-cigarette Initiators (who had used e-cigarettes exclusively; n = 414), Wave 2 Cigarette Initiators (who had used combustible cigarettes exclusively by the time of Wave 2 data collection; n = 46); and Wave 2 tobacco naïve (who reported no use of any tobacco product including cigarette, e-cigarette, traditional cigar, cigarillo, filtered cigar, pipe tobacco, hookah, snus tobacco, other smokeless tobacco, and dissolvable tobacco by the time of Wave 2 data collection; n = 5849). Participants reported their combustible cigarette ever use and/or past 30-day use at Wave 3 data collection (October, 2015 – October, 2016).

This sample selection process (see supplementary Figure S1) was necessitated by the design of the study which identified tobacco naïve youth at Wave 1, who then provided data on e-cigarette exposure at Wave 2, and also provided outcome data at Wave 3. Compared to participants who completed Waves 1 – 3 data collections (n = 7595), those who were missing from Wave 2 or Wave 3 (n = 1413) were more likely at Wave 1 to be cigarette ever users (7.6% used among completed vs 9.3% used among missing, $p<.001$) and younger (75.5% younger among completed vs 89.5% among missing, $p = .023$), but groups did not

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differ by e-cigarette ever use status (6.3% used among completed vs 7.3% used among missing) and gender (51.7% male among completed vs 51.7% male among missing).

Measures

Pre-exposure confounders.—A set of 57 pre-exposure confounders (assessed in Wave 1, 2013 – 2014) were included to adjust for confounding and selection effects related to e-cigarette use and subsequent combustible cigarette use. These variables were derived primarily from the literature on e-cigarette use and combustible cigarette use initiation, including self- or parent-report on socio-demographic (e.g., sex, race), interpersonal (e.g., depression, impulsivity, medical history), behavioral (e.g., alcohol use, drug use) and social environmental (e.g., living with smokers, parental monitoring) factors. These factors represented measures on the domains of the Host, Agent, Vector, Environment (HAVE) conceptual model, which has been validated in assessing the impact of several tobacco control policy interventions (Hyland et al., 2017). Descriptions of these confounders are presented in Supplementary Table S1.

Exposure.—Tobacco ever use was assessed in Wave 2 (2014 – 2015). Based on PATH derived variables indicating whether youth have ever used e-cigarettes, combustible cigarettes, or other tobacco variables at Wave 2. Two binary indicators were created to indicate whether a participant had ever exclusively used e-cigarettes (Question 1), or cigarettes (Question 2) at Wave 2 (1 = Yes, 0 = No). The reference group for both questions included those who remained tobacco naïve at Wave 2.

Outcome.—Combustible cigarette use initiation and past 30-day (P30D) cigarette use were assessed at Wave 3 (2015 – 2016). The outcome variable for Question 1, Wave 3 cigarette initiation, was identified by a PATH derived variable indicating whether a participant started smoking combustible cigarettes between Wave 2 and Wave 3 (1 = Yes) or not (0 = No). The outcome variable for Question 2, Wave 3 P30D cigarette use, was defined by self-reported combustible cigarette smoking in the past 30 days (1 = Yes, 0 = No).

Analytical Strategy

To rigorously control for the confounding and selection bias in the observational PATH study, we applied the entropy balancing (EB) method. EB (Hainmueller, 2012) is a multivariate reweighting method which directly adjusts the weight of each participant so that the covariate distributions in the reweighted data achieve balance. This method minimizes the entropy distance between a uniform weight (inverse of sample size) and the solution weight, which satisfies criteria including (1) each participant in the exposure group has a weight of 1 and in the unexposed group has a positive weight; and (2) all covariates must meet a pre-specified tolerance level of covariate balance (Austin, 2009; Austin & Stuart, 2015). Simulation studies (Zhao & Percival, 2016) have concluded that EB achieves similar or even superior covariate balance compared to alternatives, such as covariate balancing propensity score methods (Imai & Ratkovic, 2014) or generalized boosted methods (McCaffrey et al., 2004). Unlike the latter method that requires iterative propensity score estimation (Hainmueller, 2012; Ho et al., 2007; Stuart, 2010), EB has the principal advantage of simultaneously balancing all of the covariates.

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In this study, similar analyses were conducted for Questions 1 and 2, respectively. We included the survey weights in the model for estimating the entropy balancing weights, using pre-exposure confounders measured at Wave 1 (Ridgeway et al., 2015). Confounder balance was identified for the first two moments (i.e., mean and variance) of a distribution. An absolute standardized difference in group means less than 0.20 (i.e., Cohen's small effect size), and a ratio of variances of 0.5 to 2 indicate balance was achieved (Austin, 2009). For the outcome analysis, we conducted weighted logistic regression using the product of sampling weights from Wave 3 and entropy balancing weights to adjust for confounding, nonresponse, and sampling bias. Lastly, we conducted sensitivity analyses to evaluate the impact of potential unmeasured confounding. The e-value statistics were computed to assess the minimum strength of an association, on the odds ratio scale, that an unmeasured confounder would need to have with both the exposure and the outcome to fully explain away a specific exposure-outcome association, conditional on the measured confounders (VanderWeele & Ding, 2017). The greater an e-value (e.g., OR > 5 indicating a large effect size; Chen, Cohen & Chen, 2010), the greater the unmeasured confounding would need to be to explain away an exposure effect, hence the more robust the estimated exposure effect against unmeasured confounding.

We took a few approaches to address the missing values. Firstly, we assumed those who failed to participated Wave 2 and/or Wave 3 were ignorable. Results of the current study could be less robust if missing values were associated with individual characteristics. Among those aged 12 – 14 years at Wave 1, 81.9% completed all three data collection waves. We could not identify all those who were aged 15 at Wave 1 due to the fact that youth were categorized in two age groups (12 – 14 years and 15 – 17 years). We assume those who were aged 15 had a similar proportion of participants complete all three data collection waves. Secondly, missing data at the item level were imputed via multiple imputation. The exposure, outcome, and all confounders variables were included in the imputation model, a recommended approach in order to increase the plausibility of an ignorable missing data mechanism (Collins et al., 2001). Given the proportion of missing values at the item level were low (0% - 3.5%), 5 complete data sets were generated. The results of data analyses using imputed data provides valid estimates and standard errors under the missing at random assumption. Final results from five imputed data sets were combined using Rubin's rules (Rubin, 2004). Lastly, we incorporated the sampling weights in the EB propensity score model and outcome analysis. These sampling weights allowed us to adjust for the missing data due to nonresponse over time.

Entropy balancing was conducted using the *ebalance* Stata package.(Hainmueller & Xu, 2013). Weighted regression outcome analyses (*proc surveylogistic*) were conducted using SAS. Missing data at the item level were imputed (m = 5 replications) via chained equations using the *mice* R package (Buuren & Groothuis-Oudshoorn, 2011). Sensitivity analyses to unmeasured confounding were conducted using the *evalue* Stata package (Linden et al., 2020).

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Results

Descriptive statistics of the study sample, with and without sampling weights, are presented in Table 1. At Wave 1, participants were aged between 12 and 15 years, half were female, half were white, and all were tobacco naïve.

Among the 414 Wave 2 exclusive e-cigarette users, 46 (11.1%) started cigarette smoking, 16 (3.9%) reported P30D cigarette use, 365 (88.2%) reported no cigarette use, and 3 were missing values at Wave 3. Among the 46 Wave 2 exclusive cigarette users, 16 (34.8%) reported P30D cigarette use at Wave 3. Among the 5849 Wave 2 tobacco naïve participants, 135 (2.3%) started cigarette smoking, 52 (0.89%) reported P30D cigarette use, and 5703 (97.5%) reported no cigarette smoking at Wave 3 (See Figure S1). Descriptive statistics of the study sample stratified by tobacco use status, with and without sampling weights, are presented in Table S2.

Question 1, "*What would be the change in risk of subsequent combustible cigarette use had the user not used ENDS?"*

EB was applied to reweight the covariates in the exclusive e-cigarette exposure and tobacco naive reference groups. The mean and variance of all 57 confounding variables achieved covariate balance. Figure 1 displays the standardized mean differences (SMDs) and ratios of variances of pre-exposure confounders before and after applying the EB weights using the first imputed data set. Before applying the EB weights, the SMDs ranged from −0.61 to 0.59, and the variance ratios were between 0.44 and 6.20. After applying the EB weights, the SMDs were all close to 0 (ranged from −0.0003 to 0.004), and the variance ratios were all close to 1 (ranged from 1 to 1.003). Results suggested that balance on all covariates was achieved and thus, e-cigarette exposure and tobacco naïve groups are comparable in the EB weighted samples. Similar results were obtained from the other imputed datasets, and are not presented due to limited space but are available from the first author by request.

Results from weighted logistic regression indicated a positive association between prior e-cigarette use and subsequent combustible cigarette initiation, OR = 3.42, 95% CI = (1.99, 5.93), and P30D combustible cigarette use, OR = 2.88, 95% CI = (1.22, 6.86), in the following year (Table 2). The results of weighted logistic regression only adjusting for sample selection are also presented in Table 2 for comparison. The odds ratios with EB adjustment are smaller than those without due to removal of confounding. Results of sensitivity analyses indicated that, above and beyond the 57 measured confounders, our estimated effect could be explained away only if an unobserved confounder was associated with both exposure and outcome by large odds ratios, when assessing the association between prior e-cigarette use and subsequent combustible cigarette initiation (e-value: adjusted OR = 6.30) and P30D combustible cigarette use (e-value: adjusted OR = 5.21).

Question 2, "*What would be the change in risk of subsequent combustible cigarette use had the user not used a combustible cigarette?*"

EB was used to reweight the covariates in the exclusive cigarette exposure and tobacco naive reference groups. For the first imputed data set, before applying the EB weights, the

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SMDs ranged from −0.73 to 0.58 and the variance ratios were between 0.21 and 7.46. After applying the EB weights, the SMDs were all close to 0 (ranged from −0.0009 to 0.07), and the variance ratios were all close to 1 (ranged between 0.99 and 1.02). Results suggested that balance on all covariates was achieved and thus, exclusive cigarette exposure and tobacco naïve groups are comparable in the EB weighted samples. Similar results were obtained from the other imputed datasets and are not presented due to limited space but are available from the first author by request.

Combustible cigarette ever use was associated with nearly 26 times the risk of subsequent current combustible cigarette use, OR = 25.79, 95% CI = (9.68, 68.72), in the following year. The results of weighted logistic regression only adjusting for sample selection are also presented in Table 2 for comparison. The odds ratios with EB adjustment are smaller than those without due to removal of confounding. Results of sensitivity analyses indicated that, above and beyond the 57 measured confounders, our estimated effect could be explained away only if an unobserved confounder was associated with both exposure and outcome by very large odds ratios (e-value: adjusted OR = 51.98), when assessing the association between combustible cigarette ever use and subsequent current combustible cigarette use.

Discussion

In this study, we examined the effect of e-cigarette ever use on subsequent combustible cigarette use, using nationally representative samples of youth who were tobacco naïve at the baseline (Wave 1) of the PATH study. Ideally, an unbiased effect of prior tobacco use on subsequent behavior outcomes would be assessed via randomized controlled trials. However, because randomization to e-cigarettes or combustible cigarette smoking is not possible, we relied on the longitudinal observational data from the PATH study. We pursued statistical balance on pre-exposure confounders, mimicking the random assignment procedures. In this study, confounders, exposures, and outcome variables were selected to specify a temporal causal order. Results indicated that confounder balance was achieved after applying EB weights. Thus, we were able to estimate the de-confounded associations between initial e-cigarette use on subsequent combustible cigarette use.

We found evidence that use of e-cigarettes among tobacco-naïve youth was associated with subsequent cigarette smoking initiation and P30D smoking. This is consistent with prior reports (Chan et al., 2021; Khouja et al., 2020a; Soneji et al., 2017). We also found that tobacco initiation via cigarettes was associated with approximately a seven-fold increase in the likelihood that youth would continue to smoke combustible cigarettes. Use of any non-cigarette tobacco product, including e-cigarettes, appears to be associated with similar, increased likelihood of subsequent cigarette smoking initiation (Khouja et al., 2020), but the risk of continuing to smoke (e.g., P30D use) is considerably higher if youth start with a cigarette. This is consistent with evidence suggesting that smoking cigarettes is associated with a greater degree of dependence compared to use of other tobacco products including e-cigarettes (Strong et al., 2017). With fast, efficient delivery of nicotine and possible effects on other neurophysiological processes related to reinforcement (Hogg, 2016), cigarettes currently remain the most addictive products on the market.

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Even though findings from previous studies showed an association between e-cigarette use and subsequent uptake of combustible cigarettes among tobacco naïve youth, attention must be paid to the differences in study designs. Our results support those of Kim and Selya (2020) in that we found that the use of e-cigarettes increased the odds of subsequent cigarette initiation. However, we also found that use of e-cigarettes increased the odds of subsequent P30D cigarette use, whereas Kim and Selya (2020) did not find that the use of e-cigarettes was associated with P30D combustible cigarette smoking in their cross-sectional sample. This latter finding could be due to differences in the number of covariates adjusted for, inclusion of survey weights in our EB model, the use of EB in the present study versus logistic regression in their study, the multiple imputation of missing data, or the cross-sectional versus longitudinal samples. One difficulty with cross-sectional studies is that confounder variables, exposure, and outcome variables are all assessed at the same time. It can be difficult to determine whether an exposure has in fact affected the outcome or vice versa. In many longitudinal observational studies, the exposure and confounders are assessed at one time and the outcome is assessed at a subsequent time (Lee & Fry, 2019; Watkins et al., 2018). Such designs allow researchers to maintain the temporal order of exposure and outcome. When confounders and exposure are assessed at the same time, the "pretreatment criterion" of confounder selection may be violated (Rubin, 2008, 2009). In contrast, data from at least three time points allows researchers to maintain the temporal ordering of pre-exposure confounders, exposure, and subsequent outcome. Thus, a strength of the present study is the longitudinal design.

A recent study implemented propensity score methods to assess the relationship between exposure and outcome adjusting for their shared influence from risk factors and other confounders (Lee & Fry, 2020). The present study differs from their study in that our analysis included 57 confounders whereas theirs included 14. They noted that further research with a larger set of confounders and longitudinal data was needed (Lee & Fry, 2020).

In the current study, we carefully controlled for a larger number of confounder variables than other similar studies (Chan et al., 2021; Khouja et al., 2020). Even though a comprehensive set of covariates was considered and used in the current study, it is possible that information relevant to initiation and use was not assessed. For example, discipline problems (i.e., the frequency of participants being disciplined in school for their misbehavior) was considered in a recent study using a national probability sample of US high school youth (Kim & Selya, 2020). Unmeasured variables that are associated with e-cigarette use and cigarette initiation or P30D use may still exist, which places limits on a strict causal interpretation of the findings. A causal interpretation relies on the assumption that there are no unmeasured pre-exposure confounders that bias the effect of initial e-cigarette use on subsequent combustible cigarette use. We conducted sensitivity analyses as recommended by VanderWeele and Ding (2017) and found that our reported exposure effects were robust against unmeasured confounding because they would be explained away only if very large confounding exists and has been ignored in the PATH study.

Strengths of the current study include the use of EB weights to achieve balance in the distributions of hypothesized pre-exposure confounder variables across the participants in

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the e-cigarette use and never used groups, thereby reducing potential confounding. Results indicated that balance in covariate distributions was achieved for the first two moments (i.e., mean and variance), whereas the commonly used regression analysis would only be able to achieve confounder balance for the first moment (i.e., mean) if the same set of confounders were adjusted for. In the results, as shown in Table 2, odds ratios with EB adjustment are smaller than those using only the sampling weights because of the elimination of confounding. Unlike other propensity score estimation procedures which often involve a cumbersome iterative process of propensity score estimation, EB simultaneously balances all covariates. In addition, we included the survey sampling weights in both the EB model for estimating the weights and in the outcome model as recommended by Ridgeway et al. (2015). Another strength of our study is that we multiply imputed missing data and combined the causal effect estimates across imputations (see e.g., Coffman et al., 2020).

The current study has some limitations. This study used self-report data and only a single item or few items to assess many complex constructs which were considered as potential confounders. Hence, some measures were vulnerable to measurement error and recall bias. On one hand, the measures in the PATH study were selected based on many used in other well-established national representative studies (e.g., National Health and Nutrition Examination Survey) to assess nicotine dependence, psychological distress, health outcomes, and other measures. On the other hand, the use of a single or a few items to assess non-demographic-type information, especially measures of psychological constructs, may lead to low validity and reliability of the measures (Fisher et al., 2016). Single and multiple mis-measured confounders can lead to biased estimates of causal effects under certain circumstances (Hong et al., 2019). Thus, the results need to be interpreted with caution.

Participant's beliefs, intentions, and interpretations of survey questions may have an impact on the estimated effect in an unexpected direction. For example, youth tobacco use prevalence estimates from the household-based PATH study are generally lower than those from school-based national surveys such as National Youth Tobacco Survey (Kasza et al., 2017). More reliable and valid measurement of tobacco use behaviors may require a more intensive data collection design (e.g., ecological momentary assessment) and a more objective measure of exposure and use via biomarkers (Chang et al., 2017). Moreover, our study must be interpreted in the context of American youth aged between 12 and 17 at the time of data collection (2013 – 2016). During this period of time, information about e-cigarette products and their availability may have been different from what has occurred more recently in terms of product design and marketing practices. Hence, we recommend caution when generalizing the conclusions of the current study to populations of a different age, from a different country, or when data are collected in different time period.

It is worth noting that the measure of past 30-day cigarette use is limited in terms of implications for understanding intensity of smoking and regular use. Past 30-day cigarette use is often referred to as a measure of current use, but it combines infrequent use (e.g., one or two occasions) with more frequent and even daily use. Thus, it does not distinguish between sporadic and more regular, dependent use. Small sample sizes precluded additional analyses based on other definitions of more frequent use. For example, in the study sample,

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few youth reported smoking cigarettes on more than 20 days (12 participants) or every day (7 participants) at Wave 3. Results of this study, therefore, should be interpreted with caution.

We also acknowledge that that missing data in the study might be related to characteristics which were not considered in the study or unmeasured in the PATH study (i.e., missing not at random) and if so results may not be robust to our missing data strategies. Procedures assuming missing complete at random or missing at random may be biased if the missing mechanism is actually missing not at random. In these cases, further studies which use repeated measures and advanced modeling approaches (Xu & Blozis, 2011) may provide more robust estimates.

Conclusion and Future Directions

Use of e-cigarettes in nicotine and tobacco-naïve youth is associated with increased risk of subsequently using combustible cigarettes but the risk is an order of magnitude higher if they start with a combustible cigarette. These findings underscore the importance of preventing youth access to e-cigarettes and other tobacco products, including via enhanced age-verification methods and other technologies that might render such products difficult to use.

Future studies should focus on the factors which mediate the transition between e-cigarette and cigarette use among U.S. youth. E-cigarette use may activate cognitive process (e.g., decreased harm perceptions regarding cigarette smoking; changes in perceived social norms) that increase the risk of subsequent smoking (Soneji et al., 2017). It is important to assess whether and how cognitive factors play a role in the transition between e-cigarette and cigarette use and/or use of other existing and new tobacco products, while accounting for complex confounding. This knowledge could be used to inform and tailor educational messages that specifically target at-risk youth.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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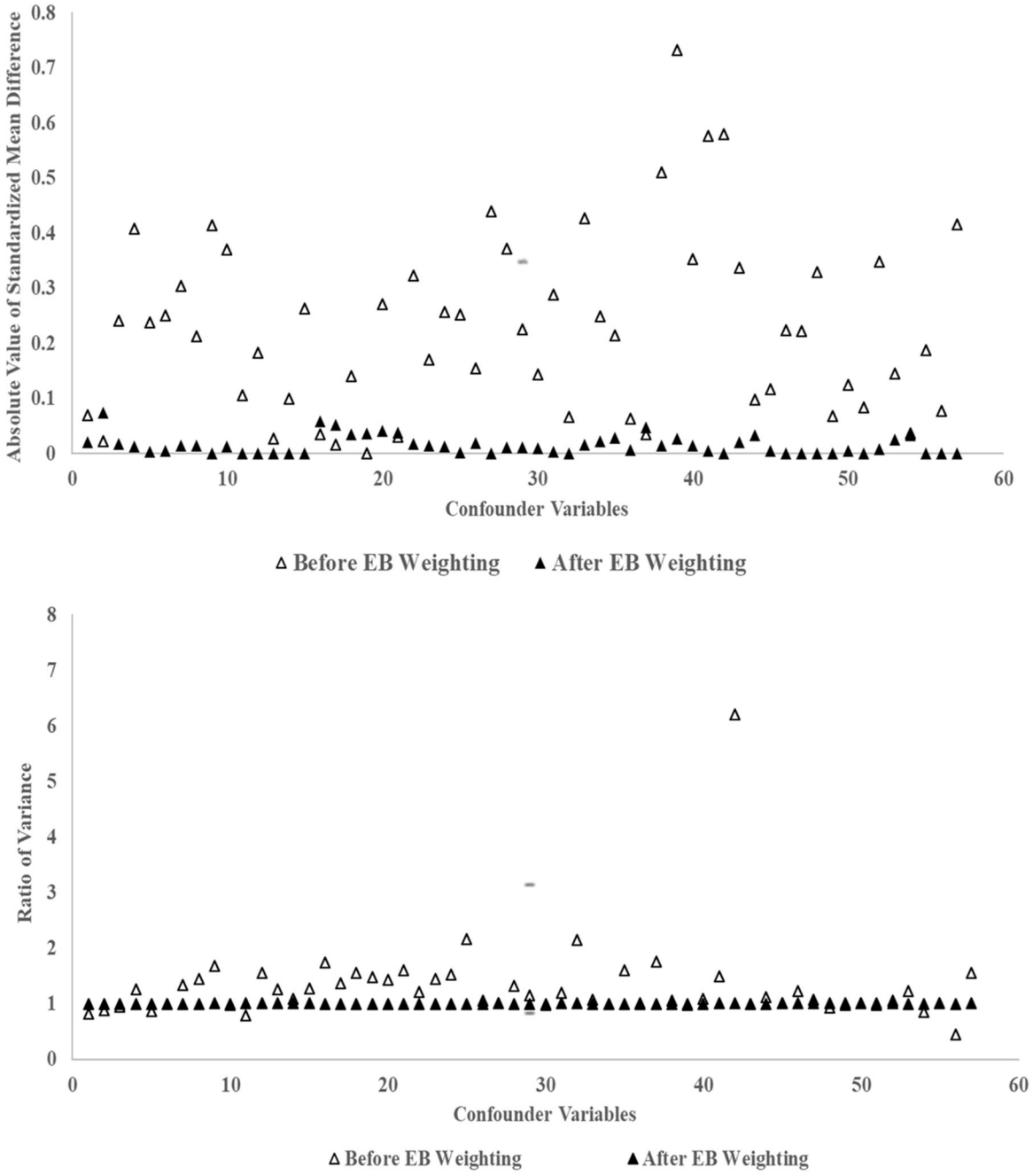


figure 1.

Confounder balance for the first imputed dataset for Question 1, What would be the change in risk of subsequent combustible cigarette use had the user not used ENDS? The top panel presents the standardized mean differences of pre-exposure confounders before and after entropy balancing. All SMDs of the 57 confounder variables after EB were close to 0. The bottom panel presents the ratios of pre-exposure confounder variance before and after entropy balancing. All ratios for the 57 confounder variables after EB were close to 1.

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Figures based on other imputed datasets for Questions 1 and 2 showed similar patterns and, hence, are not presented.

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Table 1.

Descriptive statistics of study participants at Wave 1 (N = 6309)

		Total Sample		
		n	**%**	**Weighted %**
Age (years)	12-15	6309		
Gender	Male	3243	51.4	51.2
	Female	3066	48.6	48.8
Hispanic	Hispanic	1862	29.5	22.5
	Not Hispanic	4447	70.5	77.5
Race	White	4354	69.0	70.4
	Black	1006	15.9	15.4
	Other	949	15.0	14.1
Grade	6th grade or below	583	8.5	9.2
	7th grade	1596	23.3	25.4
	8th grade	1590	24.4	25.3
	9th grade	1414	24.0	22.4
	10th grade	851	15.5	13.8
	11th grade	96	1.8	1.5
	Other *	160	2.6	2.5
Material Status	Married	4198	66.5	67.3
(Parent)	Widowed, divorced or separated,	1233	19.5	20.2
	Never married	858	13.6	12.5
Education level	Less than high school	1247	19.8	16.7
(Parent)	High school graduate or equivalent	1095	17.4	16.7
	Some college (no degree) or associates degree	1923	30.5	30.1
	Bachelor's degree	1319	20.9	23.4
	Advanced degree	690	10.9	13.2

Note.

* Not enrolled this year or last, home schooled, school not graded, 12th grade, college or vocational school.

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Table 2

The Effects of E-cigarette Ever Use on Subsequent Combustible Cigarette Use

Outcome	Sampling Weights Only			EB + Sampling Weights		
	B	SE	OR (95% CI)	B	SE	OR (95% CI)
Question 1[a]						
Cigarette Initiation	1.77	0.20	5.87 (3.97, 8.67)	1.23	0.28	3.42 (2.00, 5.93)
Past 30-day Cigarette Use	1.57	0.28	4.81 (2.66, 8.08)	1.06	0.44	2.88 (1.22, 6.86)
Question 2 [b]						
Past 30-day Cigarette Use	4.22	0.32	68.03 (36.60, 127.74)	3.25	0.49	25.79 (9.68, 68.72)

Notes. EB = Entropy balancing; B: unstandardized regression coefficient; SE = standard error; OR = odds ratio. The first set of ORs was based on a model adjusted for sampling weights only. The second set of ORs was based on a model adjusted for sampling weights and entropy balancing weights, where the entropy balancing model also used sampling weights.

[a]Question 1: *What would be the change in risk of subsequent combustible cigarette use had the user not used ENDS?*

[b]Question 2: What would be the change in risk of subsequent combustible cigarette use had the user not used a combustible cigarette?